TAIWAN SEMICONDUCTOR MANUFACTURING CO (TSMC)

NYSE: TSM (ADR, 1 ADR = 5 shares) | TWSE: 2330 | Foundry and advanced packaging for AI chips | Taiwan

Tommy Lau | 6 October 2026 | Initiation of coverage | Independent research

INITIATE AT NO CALL	**Target price:** None (no call)	**Last price:** US$485.80 (NYSE close, 5 October 2026)	**Upside:** Base value US$466 (-4%)

TSMC is the bottleneck today. At US$486, the ADR already pays for it still being one in 2028.

TSMC makes almost every advanced AI chip. Its ADR closed at US$485.80 on 5 October 2026, up 74% in a year and within two dollars of its 52-week high. On 2 October I argued that, on today's horizon, TSMC's factories bind before the grid does, and that over the longer run the grid is still the slower clock.

The near horizon is holding. TSMC shipped 17% more wafers in the second quarter and earned about 17% more on each, at a 67.7% gross margin, almost twelve points above its own through-the-cycle floor. July and August revenue put the third quarter on course to beat the top of the guide. The price knows it: at 20 times, it needs a 2028 operating margin of about 58%, the scarcity margin, just as the new fabs land.

My call: NO CALL, medium conviction. My base case, close to TSMC's own five-year plan, is worth US$466, down 4%. The bear case, in which the grid becomes the slower clock, is US$278 (down 43%), against a bull case of US$658 (up 36%). I would revisit below about US$400.

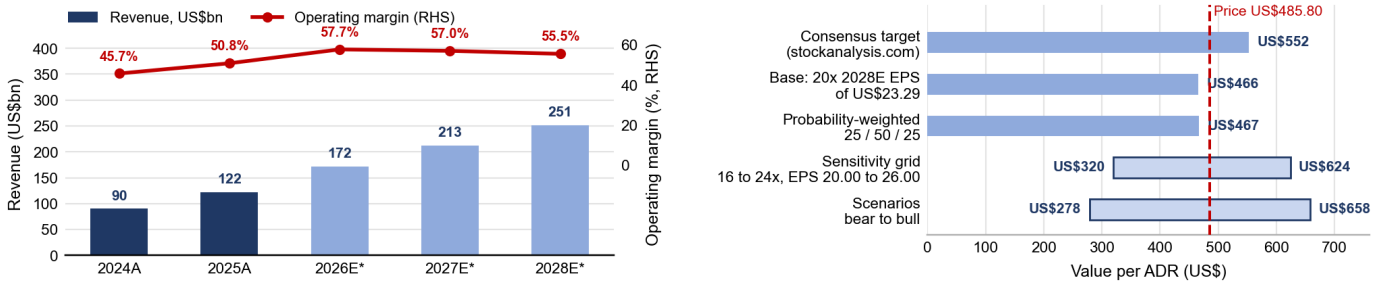
Stock data
LAST PRICE, ADR **US$485.80, NYSE close 5 October 2026; 52-week range US$266.82 to 487.47**
LAST PRICE, TAIPEI **NT$2,585, TWSE close 6 Oct 2026 (2330.TW)**
ADR PREMIUM TO TAIPEI **19% at NT$31.75 per US$ (5 shares per ADR)**
MARKET VALUE **US$2.52tn at the ADR price, US$2.11tn at Taipei's; 25,932m shares, 30 Jun 2026**
CASH LESS LONG-TERM DEBT, 30 JUN 2026 **About US$84bn (cash and securities NT$3,518bn less long-term debt NT$864bn)**
P/E ON MY ESTIMATES **28.9x 2026E, 24.0x 2027E, 20.9x 2028E**
CONSENSUS **2027 EPS US$21.33 per ADR (MarketBeat); forward P/E 20.7x, target US$552 (stockanalysis.com)**
Q3 2026 GUIDE (16 JUL) **Revenue US$44.6 to 45.8bn; gross margin 65 to 67%; operating margin 56 to 58%, at NT$32**
NEXT RESULTS **Q3 2026: Thursday 15 Oct 2026, 2pm Taiwan time**

Key financials

US$ billion unless stated	2024A	2025A	2026E*	2027E*	2028E*
Revenue	90.08	122.42	171.50	212.66	250.94
Revenue growth (%)	n.a.	36	40	24	18
Gross margin (%)	56.1	59.9	n.m.	n.m.	n.m.
Operating margin (%)	45.7	50.8	57.7	57.0	55.5
EPS per ordinary share (NT$)	45.25	66.25			
EPS per ADR (US$)	7.04	10.65	16.81	20.25	23.29
P/E at US$485.80 (x)	69.0	45.6	28.9	24.0	20.9

**2026E to 2028E are my own estimates. 2026E uses the reported first half (US$7.80 per ADR) plus my third and fourth quarters (revenue US$46.4 and 49.0bn; operating margin 57% and 56%). EPS = revenue x (operating margin + 2.5% non-operating) x (1 - 17% tax and minorities) x 5 / 25.932bn shares. I model operating, not gross, margin ahead (n.m.). 2024A and 2025A from TSMC's 4Q25 presentation; per-ADR EPS summed from quarterly releases. My 2027E sits 5% below the MarketBeat consensus of US$21.33.*

Two charts: the margin path I expect, and a valuation that brackets the price



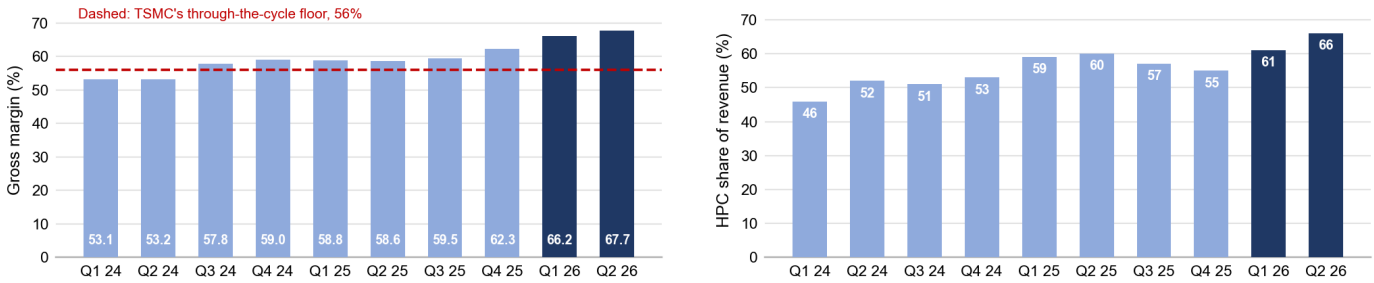
Left: revenue and operating margin, 2024A to 2028E; 2024A and 2025A per TSMC, the rest my estimates. Right: value per ADR from the base case, the probability-weighted cases, the sensitivity grid and the bear to bull range, against the US$485.80 price (red) and the stockanalysis.com average analyst target, retrieved 6 October 2026.

The thesis: a kitchen with a queue cooks more and charges more

Go back to the shared kitchen from the October piece. Every famous restaurant designs its own dishes, and all of them send the cooking to one kitchen. When demand surges, the queue at that kitchen sets how fast every restaurant grows. A kitchen with a queue does two things: it cooks more meals, and it charges more for each one. TSMC is doing both.

More wafers. TSMC shipped 4.34 million twelve-inch-equivalent wafers in the second quarter of 2026, 17% more than a year earlier, from fabs that were already full. **More per wafer.** Revenue in Taiwan dollars rose 36%, so revenue per wafer rose about 17% too, part mix towards the newest processes and part price. **More of each sale kept.** Gross margin reached 67.7%, against 53.2% two years earlier and TSMC's own target of 56% and higher through the cycle.

Where the pull comes from. High-performance computing, which holds AI accelerators, was 66% of revenue in the second quarter, against 46% at the start of 2024. AI accelerators alone were in the high teens as a share of 2025 revenue, and TSMC expects them to grow at a mid to high 50s per cent rate a year from 2024 to 2029. **The packaging pinch.** In July C.C. Wei said TSMC's packaging capacity is so tight that it is limiting customers' growth. That is the near horizon of the claim in the company's own words.



Left: gross margin by quarter, Q1 2024 to Q2 2026, against TSMC's through-the-cycle floor of 56% (dashed). Right: high-performance computing as a share of revenue. Source: TSMC quarterly results releases and presentations, 18 April 2024 to 16 July 2026; long-term margin from the 4Q25 presentation, 15 January 2026.

The newest evidence. TSMC publishes revenue monthly, and the third quarter is two thirds reported. July and August came to NT$982.4bn. At TSMC's assumed NT$32 per US$, the US$44.6 to 45.8bn guide needs only NT$445 to 483bn in September; August alone was NT$514.8bn. Unless September falls sharply, the quarter lands at or above the top of the guide.

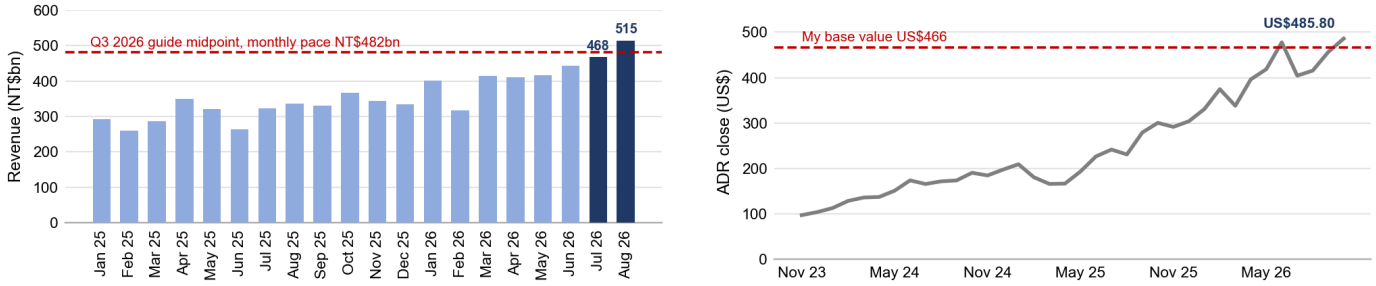
Spending buys later supply. On 15 January Wei said new spending adds almost nothing to output this year, speaking of the original 2026 capital budget of US$52 to 56bn. On 16 July TSMC raised that budget to US$60 to 64bn. The larger budget makes the point stronger: spending decided in 2026 buys supply for 2028 and 2029.

Variant perception: what the market prices in, and where I differ

At US$485.80 the ADR values TSMC at about US$2.52tn, with roughly US$84bn of net cash. That is 28.9 times my 2026 estimate of US$16.81 per ADR and 24.0 times my 2027 estimate of US$20.25; on MarketBeat's 2027 consensus of US$21.33 it is 22.8 times. The ADR also trades at about 19% above five Taipei shares, which closed at NT$2,585 on 6 October.

What the price needs. By October 2027 the market will price 2028. At 20 times, close to TSMC's forward multiple today, US$485.80 needs 2028 earnings of US$24.29 per ADR. On my 2028 revenue of US$251bn, that needs an operating margin of

about 58%, against my 57.7% for 2026. The market is paying for today's scarcity margin to last into 2028, the year the October piece named as the test of the longer horizon, when the US$60 to 64bn of 2026 spending starts producing wafers.



Left: monthly revenue, January 2025 to August 2026, against the monthly pace of the Q3 guide midpoint at NT$32 per US$ (TSMC monthly revenue reports on Form 6-K). Right: TSM ADR month-end close, November 2023 to the 5 October 2026 close, not adjusted for dividends (Yahoo Finance, retrieved 6 October 2026), against my base value.

Where I differ. I agree on the near horizon. I differ on what happens when the new capacity lands, because three physical things push margin down in 2028 whatever demand does. **Depreciation:** capital spending rose from NT$956bn in 2024 to NT$1,272bn in 2025, and the first half of 2026 alone was NT$847bn; that charge arrives with the output. **Distance from the cluster:** TSMC guides that overseas fabs cut gross margin by two to three points at first, widening to three to four, and Arizona's second fab is due in volume in the second half of 2027. **The newest process:** TSMC expects the 2 nanometre ramp to cut gross margin by about three to four points in the second half of 2026.

TSMC has offset all of this with price, which works only while customers queue. That is where the longer horizon comes in. If the grid is the slower clock, the capacity arriving in 2028 meets customers whose data centres are waiting for power: the queue shortens just as the kitchen gets bigger. TSMC's own plan points the same way. Revenue growth approaching 25% a year from 2024 to 2029 implies about US$275bn in 2029; with 2026 near US$172bn on my estimate, that leaves about 17% a year for 2027 to 2029. My base case follows that plan, not a bust.

Valuation, with the working

I value TSMC on earnings per ADR, because it sells mostly in US dollars, carries net cash and pays a steady, rising dividend (NT$24 a share in 2026). The horizon is twelve months, to October 2027, so the year I value is 2028. Non-operating items jumped to NT$95.8bn in the second quarter from NT$28.8bn in the first, the quarter TSMC sold part of its stake in Vanguard International. I treat the jump as non-recurring: on my working it added about US$0.34 to second quarter EPS per ADR. Ahead, non-operating income is 2.5% of revenue, its first quarter level.

The multiple. I use 20 times 2028 earnings for the base case. That is TSMC's own forward multiple today (20.7x), below the 22 to 24 times of GlobalFoundries and UMC, and it assumes no rerating either way. 20 x US$23.29 = US$466.

Scenarios: three cases, twelve months out

Case	What happens	2028 EPS	Multiple	Value	Change	Weight
Bear	The grid binds first: growth 15% then 5%; operating margin 55% in 2027, 50% in 2028	US$17.40	16x	US$278	down 43%	25%
Base	TSMC's own plan: growth 24% then 18%; margin 57% then 55.5%	US$23.29	20x	US$466	down 4%	50%
Bull	The factory stays the bottleneck: growth 30% then 25%; margin holds at 59%	US$27.43	24x	US$658	up 36%	25%
Weighted	**25 / 50 / 25**			**about US$467**	**down 4%**	

All cases start from my 2026E revenue of US$171.5bn and use the same earnings bridge. Values are per ADR, unrounded in the model. The downside in the bear case is larger than the upside in the bull case.

Sensitivity: 2028 EPS against the multiple

2028 EPS per ADR	16x	20x	24x
US$20.00	US$320	US$400	US$480
US$23.29	US$373	**US$466**	US$559
US$26.00	US$416	US$520	US$624

Only 3 of the nine cells sit above today's price. To make money from here, TSMC needs either earnings above my base case or a higher multiple than my base case uses. Every figure is live in the accompanying model, so a change to growth, margin or the multiple moves the value.

Peer landscape

Company	Ticker	Forward P/E	What they make
TSMC	NYSE: TSM	20.7x	Foundry: makes chips others design; CoWoS advanced packaging
GlobalFoundries	Nasdaq: GFS	22.4x	Foundry, mature and specialty processes
United Microelectronics	NYSE: UMC	24.0x	Foundry, mature processes; makes interposers
Intel	Nasdaq: INTC	68.8x	Designs and makes its own chips; Intel Foundry for outside customers
ASE Technology	NYSE: ASX	31.4x	Chip packaging and test, including advanced packaging
Samsung Electronics	KRX: 005930	4.5x	Memory, including stacked HBM; Samsung Foundry
Nvidia (customer)	Nasdaq: NVDA	19.8x	Designs AI accelerators; has them made at TSMC

Forward P/E from stockanalysis.com, retrieved 6 October 2026. Intel's and Samsung's multiples are distorted by recovery and memory cycles. Nvidia is TSMC's largest AI customer, shown for context. Multiples only; no view on the peers' shares.

Catalysts

When	Event	What I watch
Around 10 Oct 2026 (date not confirmed)	September 2026 revenue	Whether Q3 lands above the US$45.8bn top of the guide
15 Oct 2026, 2pm Taiwan time	Q3 2026 results and call	Gross margin against 65 to 67% with the 2nm ramp; Q4 guide; whether packaging is still called tight; any 2027 capex signal
January 2027	Q4 2026 results	2027 capital budget and revenue outlook; any change to the 56% through-the-cycle margin floor
Second half of 2027	Arizona second fab in volume	Overseas margin dilution moving from 2 to 3 points towards 3 to 4
Monthly	Revenue reports	High-performance computing share and the run-rate against each guide

Risks: what would hurt the view

The factory stays the bottleneck for longer. The bull case and the main risk to a no call. If AI demand outruns the new fabs into 2028 and the grid keeps up, the scarcity margin survives. A raised margin floor would be the clearest sign.

The grid binds first, and sooner. The bear case: TSMC's tightness eases while data centres with chips on order still wait for power, which is the near horizon of my claim breaking. Watch for packaging called balanced, or high-performance computing slipping as a share of revenue.

The exchange rate and the ADR premium. TSMC reports in Taiwan dollars, sells mostly in US dollars and guides margin at a set rate. A stronger Taiwan dollar lowers margin and earnings per ADR, and the ADR's 19% premium to the Taipei shares can narrow even if the business does well.

Taiwan's electricity. In January Wei said his first worry on power was Taiwan's own supply. A fab is a building waiting on power too. **Rival foundries.** Intel Foundry and Samsung Foundry want TSMC's AI customers. I found no sign of a large AI accelerator moving away, but packaging is where a second source is easiest.

What would change my mind

Into a call: a price below about US$400, which would put my base case about 16% above it; or third quarter results showing gross margin at or above the 67% top of the guide despite the 2 nanometre ramp; or TSMC raising its through-the-cycle margin floor above 56%.

The call is wrong if full year 2027 revenue grows 30% or more in US dollars with an operating margin of 58% or more, or TSMC raises its through-the-cycle gross margin floor above 56%, or the ADR trades above US$560 by October 2027 without either.

Conclusion

The October piece argued that, today, TSMC's factories bind before the grid does. A quarter of TSMC's own numbers says that is right: more wafers, more per wafer, and a margin almost twelve points above its own floor. The shares know it. At US$485.80 the ADR needs the scarcity margin to survive into 2028, when new capacity lands and the longer horizon of my claim begins to be tested. My call: NO CALL, medium conviction. Revisit below about US$400.

Appendix: forecast assumptions

Assumption	2026E	2027E	2028E	Basis
Revenue growth, US$	40%	24%	18%	2026 on TSMC's guide and monthly revenue; then TSMC's 2024 to 2029 plan
Operating margin	57.7%	57.0%	55.5%	H2 2026 guide 56 to 58%; then depreciation, overseas and 2nm dilution
Non-operating income, % of revenue	2.5%	2.5%	2.5%	Q1 2026 level; Q2 2026 jump excluded
Tax and minorities, % of pre-tax	17%	17%	17%	2025: 15.9%; first half 2026: 17.5%
Target multiple			20x	TSMC's forward P/E today

Full workings, with live formulas, in the accompanying model, 2026-10-06_TSMC_Model.xlsx.

Sources

TSMC quarterly results releases and presentations furnished on Form 6-K, 18 April 2024 to 16 July 2026, for revenue, margins, EPS per share and per ADR, technology and platform mix, wafer shipments, non-operating items, cash, debt, share count, capital expenditures and the Q3 2026 guide. TSMC 4Q25 presentation, 15 January 2026, for 2024 and 2025 annual figures and its 2024 to 2029 targets. TSMC monthly revenue reports on Form 6-K, 10 February 2025 to 10 September 2026. TSMC earnings call transcripts of 15 January and 16 July 2026. TSMC Form 6-K of 15 May 2026 on the Vanguard International share sale. TSMC investor relations for the 15 October 2026 results date. Consensus from MarketBeat and stockanalysis.com, and peer multiples from stockanalysis.com, retrieved 6 October 2026. ADR, Taipei share and exchange rate prices from Yahoo Finance, retrieved 6 October 2026. Estimates for 2026 to 2028 are the author's own.

About this note

I wrote this note and built the accompanying model independently. It is not a product of any bank or broker. It tests a claim from my journal piece of 2 October 2026, thephysicallayer.fyi/journal/tsmc-says-it-is-the-bottleneck; all notes and their track record are at thephysicallayer.fyi/research. Figures from company filings are labelled as such; derived figures are marked; the forecasts and view are my own estimates and my own view. I hold no position in TSMC. Personal research, not investment advice.

Tommy Lau | The Physical Layer | thephysicallayer.fyi